

Comparative Analysis of Spatial, Edge-Preserving and Transform-Based Filters for Medical Image Enhancement

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Abstract

Magnetic Resonance Imaging (MRI), Computed Tomography (CT) and ultrasound are commonly used medical imaging techniques for diagnosis and treatment planning. But images acquired from such systems are commonly contaminated with noise during acquisition, transmission or reconstruction. Denoising is therefore crucial for improving image quality without compromising critical anatomical features and diagnostic information.

This study presents a detailed comparative analysis of various denoising techniques such as classical spatial filters, edge preserving methods and transform domain approaches. This includes Mean, Median, Gaussian, Wiener, Bilateral, and Anisotropic Diffusion filters, as well as advanced approaches such as Non-Local Means, Wavelet Transform, Curvelet Transform, Moving Average, Exponential Moving Average, and a hybrid Median-Mean filtering strategy. The performance of these techniques was quantitatively evaluated using image quality metrics, including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Signal-to-Noise Ratio (SNR).

Experimental results demonstrate that transform domain methods and Non-Local Means filtering achieve superior structural preservation and higher perceptual quality, while classical filters provide lower computational complexity and faster processing. The findings of this study offer valuable insights for selecting suitable denoising techniques for different medical imaging modalities and clinical applications, contributing to improved reliability in medical image analysis and computer-aided diagnosis systems.

Keywords: Medical Image Denoising, Median Filter, Gaussian Filter, Anisotropic Diffusion, Wavelet Transform, Curvelet Transform, PSNR, SSIM, Medical Imaging

1. Introduction

Medical imaging is essential for diagnosing diseases, planning surgeries, and monitoring treatments. However, medical images often contain different types of noise, including Gaussian noise, salt-and-pepper noise, and speckle noise[1]. Noisy images create a significant problem for healthcare professionals, including radiologists, doctors, and specialists. Noise is an unwanted and random signal that reduces image quality and may hide important diagnostic information, which can impact the accurate detection of diseases such as tumors[2].

Over the years, many denoising techniques have been proposed, from traditional linear smoothing filters to advanced transform-domain and non-local approaches. These techniques aim to improve image quality while minimizing the loss of critical information[3][4][5][6]. However, selecting an

appropriate denoising technique remains a difficult task because different methods exhibit varying performance based on the type of noise, imaging modality, and computational constraints. For example, spatial filters are computationally simple but can cause edge blurring, whereas transform-domain and non-local approaches provide better structural preservation at the cost of higher computational complexity.

Thus, a systematic comparison of different denoising techniques is necessary to assess their effectiveness for various medical imaging methods.

This study presents a detailed comparison of various denoising techniques used on CT, MRI, and Ultrasound images. The techniques are divided into three major groups: spatial domain filters, edge-preserving filters, and transform-based methods. The performance of these techniques is analyzed using standard image quality metrics such as Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Signal-to-Noise Ratio (SNR). The goal is to identify the most effective denoising strategies that offer a good balance between noise reduction, structure preservation, and computational efficiency in medical imaging. The sections are organized as follows: Section 2 presents the literature review, followed by Section 3 on the dataset, Section 4 on the methodology, Section 5 outlining the evaluation metrics, Section 6 discussing the results, Section 7 covering applications, and Section 8 concluding the study.

2. Related Works

Medical image denoising has received a lot of attention because it helps improve diagnostic accuracy and keeps important structural details in medical images. Many techniques, including filtering, transform-based methods, and learning-based approaches, have been developed to remove noise while preserving key anatomical information.

Sagheer and George [7] carried out an in-depth review of denoising algorithms used across various imaging modalities like MRI, CT, PET, and ultrasound. Their work organized denoising methods into three categories: spatial filtering, transform-domain methods, and statistical techniques. They found that transform-based methods, such as wavelet transforms, are more effective at suppressing noise while retaining structural details.

Kaur and Dong [8] provided a thorough survey on medical image denoising techniques. They discussed different noise models that affect medical images, such as Gaussian, Poisson, and speckle noise. They stressed the need to choose the right denoising methods based on the imaging type and noise characteristics. Their research also pointed out that machine learning and deep learning techniques are becoming increasingly important for improving denoising performance.

Buades et al. [9] introduced the Non-Local Means (NLM) algorithm, which uses the redundancy found in natural images to eliminate noise while keeping fine structural details. This method calculates the weighted average of pixels based on the similarity of neighboring areas in the image. Experimental results showed that NLM significantly outperforms traditional methods like Gaussian smoothing and anisotropic diffusion in preserving edges and textures in MRI images.

Elhoseny et al. [10] proposed an optimized bilateral filtering technique for medical image denoising in combination with a convolutional neural network (CNN) classifier. They enhanced the bilateral filter using bio-inspired algorithms to improve noise removal while maintaining edges. The results indicated better Peak Signal-to-Noise Ratio (PSNR) and higher classification accuracy compared to standard filtering methods.

Vilimek et al. [11] conducted a comparative study of wavelet transform-based filtering methods for ultrasound image denoising. Their findings showed that wavelet-based techniques effectively reduce speckle noise while preserving anatomical features and edges. They confirmed that the multi-resolution analysis offered by wavelet transforms enhances denoising performance compared to regular spatial filters.

Fan et al. [12] reviewed both traditional and modern image denoising techniques, covering spatial filtering, transform-based methods, and statistical models. The authors highlighted that no single denoising method is best for all types of noise or imaging modalities. Instead, choosing the right

technique depends on the noise characteristics and specific application needs. The comparative analysis of the various approaches is shown in Table 1.

Table 1: Comparative analysis of various approaches

Author	Techniques Covered	Modalities Used	Metrics Used	Comparative Study	Key Limitation
Sagheer & George [7]	Wavelet Transform	General medical images		No multi-method comparison	Focuses only on wavelets
Kaur & Dong [8]	Multiple denoising methods (survey)	General		No practical comparison	Theoretical overview only
Buades et al. [9]	Non-Local Means	General	PSNR	Single method	No comparison with other filters
Elloisky et al. [10]	Bilateral Filter	Medical images	PSNR	Limited comparison	Focus on one filter
Vimal et al. [11]	Wavelet Transform	Medical images	Basic metrics	Limited scope	No cross-category comparison
Fang et al. [12]	Spatial + statistical methods	General		No deep evaluation	Lacks multi-metric validation

The literature shows that many techniques have been developed for medical image denoising. While traditional spatial filters are computationally efficient, they may blur important edges. On the other hand, transform-domain and non-local methods provide better structural preservation. Recent advances in deep learning enhance denoising performance but also lead to increased computational complexity. A systematic comparative analysis of both classical and advanced filtering techniques is essential for finding the best denoising methods for various medical imaging modalities like CT, MRI, and ultrasound.

Contributions of the study are as follows:

The primary objective of this study is to perform a comprehensive comparative analysis of various denoising filters applied to medical images and to evaluate their effectiveness in preserving diagnostic information.

The specific objectives are:

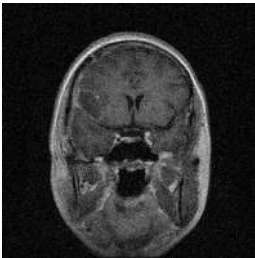
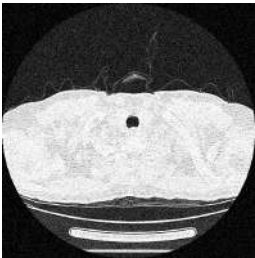
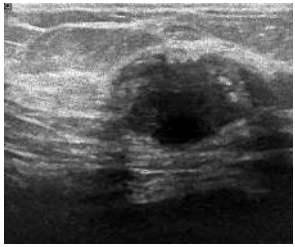
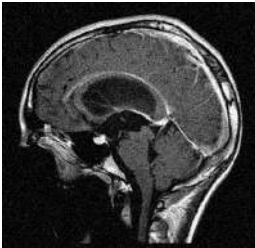
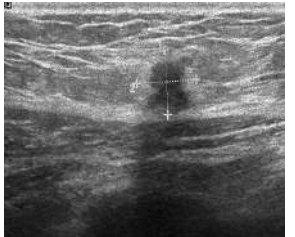
1. To analyze different categories of denoising techniques, including spatial domain filters, edge-preserving filters and transform-based methods.
2. To evaluate the performance of each filter under different noise conditions such as Gaussian noise, salt-and-pepper noise and speckle noise.

3. To quantitatively compare denoising performance using standard evaluation metrics including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) and Signal-to-Noise Ratio (SNR).
4. To assess edge preservation and structural integrity after filtering, particularly in clinically significant regions.
5. To analyze computational complexity and processing time of each technique for practical medical applications.
6. To determine the suitability of each filtering method for specific medical imaging modalities such as MRI, CT and ultrasound.
7. To identify the most effective denoising strategy that balances noise suppression, structural preservation and computational efficiency.

3. Dataset

Experiments were conducted on medical images including MRI brain scans[13], CT head images[14] and ultrasound images[15] as shown in Fig 1. The research used three publicly available medical imaging datasets consisting of Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and ultrasound images. These datasets were selected to evaluate the proposed method across different medical imaging modalities.

The MRI dataset was obtained from the Brain Tumor MRI Dataset available on Kaggle, which contains brain MRI scans of different tumor types along with non-tumor cases. The CT dataset used in this study is the COVID-19 CT Scan Lesion Segmentation Dataset, which includes CT scans with lesion annotations related to COVID-19 infection. The ultrasound dataset was taken from the Breast Ultrasound Images Dataset, which contains breast ultrasound images categorized into benign, malignant, and normal classes. Sample images from the dataset are shown in Fig 1.

MRI	CT	ULTRASOUND
		
		

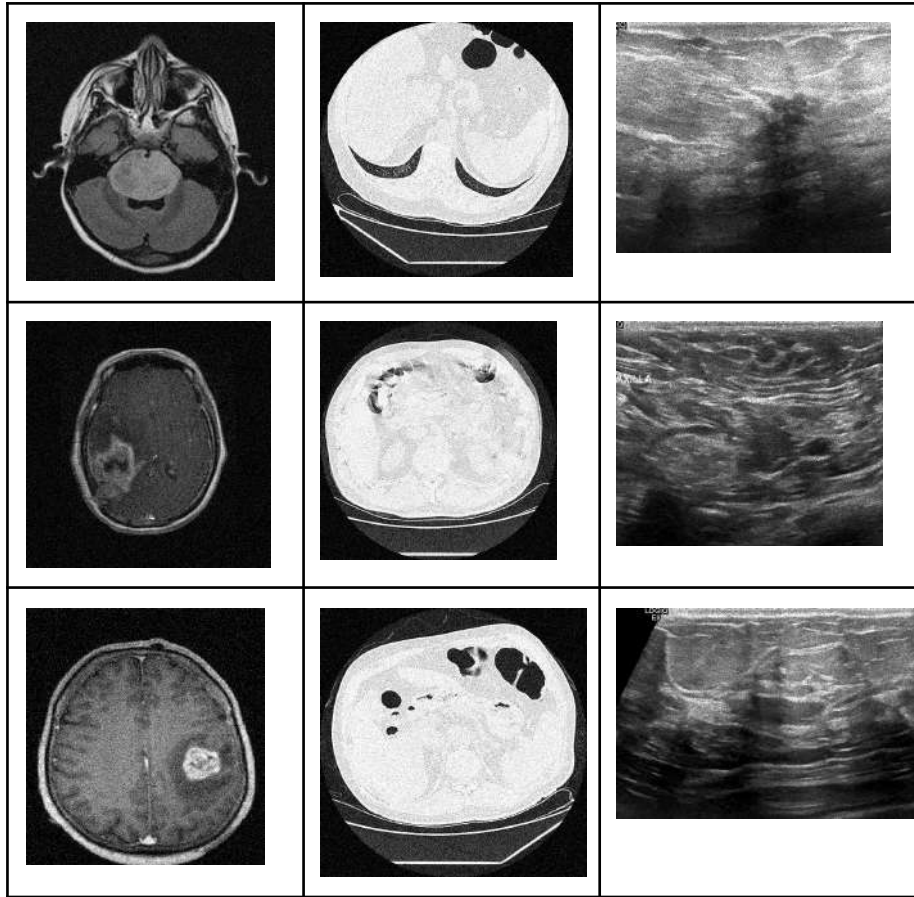


Fig 1: Sample images from three modalities- MRI, CT and Ultrasound

4. Methodology

The details of the proposed methodology consists of various steps:

1. Noise Modeling
2. Filtering Techniques

These steps are designed to evaluate the effectiveness of various denoising approaches.

Fig 2 depicts the framework of the proposed approach for comparative analysis of spatial, edge-preserving, and transform-based denoising filters for medical image enhancement.

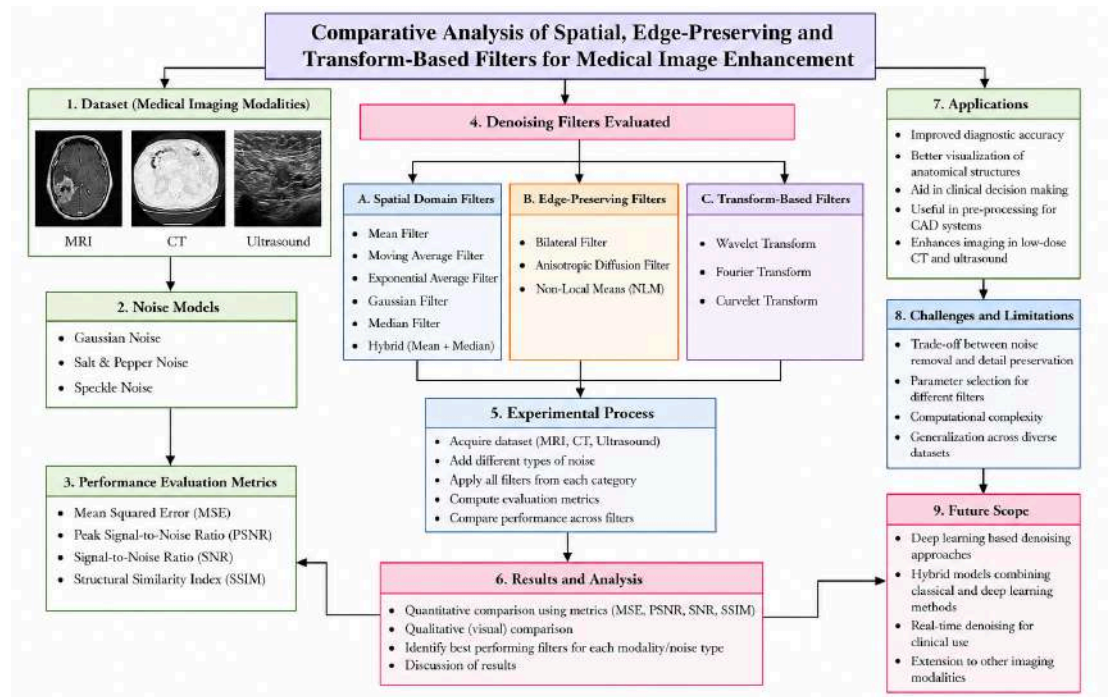


Fig 2: Framework of the proposed comparative analysis of denoising filters for medical images.

4.1 Noise Modeling

To simulate realistic imaging conditions, three types of noise were artificially introduced:

- Gaussian noise ($\sigma = 0.01-0.1$): Gaussian noise is a statistical noise model commonly used in medical imaging systems due to electronic fluctuations during image acquisition. In this model, the noise values follow a normal (Gaussian) distribution characterized by a mean and variance. It affects almost every pixel in the image and results in slight intensity variations across the image. Gaussian noise is widely used in experiments to simulate realistic degradation in MRI and CT images.
- Salt-and-pepper noise (density 0.01-0.05): Salt-and-pepper noise, also referred to as impulse noise, appears as randomly distributed bright and dark pixels in an image. It is typically caused by errors during image transmission, sensor malfunction, or faulty memory locations. In this noise model, some pixels are replaced with minimum intensity values (black) or maximum intensity values (white), which can significantly distort image details if not properly removed.
- Speckle noise: Speckle noise is a multiplicative noise commonly found in ultrasound imaging systems. It occurs due to the interference of reflected signals from multiple scatterers within biological tissues. Unlike additive noise, speckle noise depends on the pixel intensity and often produces a granular appearance that reduces image contrast and obscures structural boundaries.

4.2 Filtering Techniques

To mitigate the effects of noise, multiple filtering techniques were employed and categorized into spatial domain filters, edge-preserving filters, and frequency/transform-based methods.

4.2.1 Spatial Domain Filters

These filters were applied to MRI, CT, and ultrasound images to evaluate their effectiveness in suppressing noise while preserving structural information.

1. Mean Filter[16]: The mean filter is a linear smoothing technique used to reduce random noise by replacing each pixel value with the average intensity of its neighboring pixels within a predefined window. This method effectively suppresses small fluctuations in intensity; however, it may blur important image edges and fine anatomical structures.

2. Moving Average Filter[17]: The moving average filter is a smoothing method that computes the average of pixel intensities over a sliding window across the image. As the window moves from one pixel location to another, the filter continuously updates the average value, producing a smoother image. This technique is simple and computationally efficient but may reduce image sharpness when applied to regions containing edges.
3. Exponential Moving Average[18]: The exponential moving average filter assigns exponentially decreasing weights to older pixel values while giving higher importance to more recent observations. This weighted averaging approach allows the filter to smooth noise while retaining more information from nearby pixels compared to simple averaging methods.
4. Gaussian Filter[16][19]: The Gaussian filter is a linear smoothing filter that uses a Gaussian-shaped kernel to perform weighted averaging of neighboring pixels. Pixels closer to the center of the kernel receive higher weights than those farther away. This property helps reduce noise while preserving general image structure better than simple mean filtering.
5. Median Filter[16][19]: The median filter is a non-linear filtering technique commonly used for removing impulse noise such as salt-and-pepper noise. Instead of averaging pixel values, the filter replaces each pixel with the median value of its neighborhood. This approach effectively removes outlier pixels while maintaining edge information.
6. Hybrid (Median + Mean): The hybrid median-mean filter combines the advantages of both median and mean filtering techniques. In this approach, median filtering is first applied to remove impulse noise, followed by mean filtering to smooth remaining intensity variations. This combined strategy improves noise suppression while maintaining structural information.

4.2.2 Edge-Preserving Filters

The following edge-preserving filters were applied to MRI, CT, and ultrasound images to assess their ability to reduce noise while maintaining anatomical boundaries.

1. Bilateral Filter[19]: The bilateral filter is an edge-preserving smoothing technique that considers both spatial proximity and intensity similarity when averaging pixel values. Unlike traditional filters, it reduces noise while maintaining sharp edges because pixels with significantly different intensities contribute less to the averaging process.
2. Anisotropic Diffusion[20]: Anisotropic diffusion is an iterative filtering process designed to smooth homogeneous regions of an image while preserving edge boundaries. The method models the diffusion of pixel intensities over time, allowing diffusion within regions of similar intensity while restricting diffusion across edges. This selective smoothing improves image quality without significant loss of structural details.
3. Non-Local Means Filter[21]: The Non-Local Means (NLM) filter removes noise by exploiting the redundancy present in natural images. Instead of relying only on local neighborhoods, this technique compares larger patches throughout the image and computes weighted averages based on similarity between patches. As a result, it effectively suppresses noise while preserving textures and fine details.

4.2.3 Frequency and Transform-Based Methods

These methods were implemented on MRI, CT, and ultrasound images to analyze their performance in multi-scale noise reduction and detail preservation.

1. Wiener Filter[16]: The Wiener filter is a statistical filtering technique that minimizes the mean squared error between the original and restored images. It adapts its behavior based on local image variance and noise characteristics, making it effective for reducing Gaussian noise while preserving image structures.
2. Wavelet Transform[22]: The wavelet transform is a multi-resolution analysis technique that decomposes an image into different frequency components at various scales. Noise can be reduced by applying thresholding to high-frequency coefficients while retaining significant structural information. Wavelet-based denoising is particularly effective in preserving edges and fine details in medical images.
3. Curvelet Transform[23]: The curvelet transform is an advanced multi-scale transform designed to efficiently represent edges and curved structures in images. Unlike wavelets, curvelets capture directional information and elongated features more effectively, making them well suited for medical images containing curved anatomical boundaries.

5. Evaluation Metrics

The following evaluation metrics were used to quantitatively assess the performance of the proposed denoising methods.

- Mean Squared Error (MSE)[24]: Mean Squared Error measures the average squared difference between the original image and the denoised image. Lower MSE values indicate better restoration quality because the processed image is closer to the original.

$$MSE = \frac{1}{n} \sum (y_i - p_i)^2$$

where y_i is the i th observed value, p_i is the corresponding predicted value for y_i , and n is the number of observations. The $\sum$ indicates that a summation is performed over all values of i .

- Peak Signal-to-Noise Ratio (PSNR)[24]: PSNR is a widely used metric for evaluating image restoration quality. It is derived from the Mean Squared Error and represents the ratio between the maximum possible signal intensity and the noise level. Higher PSNR values generally indicate better image quality.

$$PSNR = 10 \log_{10} (MAX_i^2 / MSE)$$

where MAX_i = maximum possible pixel value (e.g., 255 for 8-bit images)

And MSE = mean square error

- Structural Similarity Index (SSIM)[25]: The Structural Similarity Index evaluates image quality based on perceived changes in structural information, brightness, and contrast. SSIM focuses on how similar the structures of two images are, making it particularly useful for medical imaging evaluation.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

where μ_x, μ_y = mean intensities

σ_x^2, σ_y^2 = variances

σ_{xy} = covariance

C_1, C_2 = constants

- Signal-to-Noise Ratio (SNR)[1]: Signal-to-Noise Ratio measures the strength of the desired signal relative to the background noise present in an image. A higher SNR value indicates that the image contains less noise and retains more useful information.

$$SNR = 10 \log_{10} (P_{\text{signal}} / P_{\text{noise}})$$

where P_{signal} = signal power and P_{noise} = noise power

6. Results/Findings

The effectiveness of the applied denoising techniques was evaluated on three medical imaging modalities: Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and Ultrasound images. These modalities exhibit different noise characteristics due to their distinct acquisition mechanisms. CT images typically contain Gaussian noise introduced during photon detection and electronic signal processing. MRI images often suffer from Rician or Gaussian noise due to magnetic field fluctuations and receiver electronics. Ultrasound images are predominantly affected by speckle noise, which results from the constructive and destructive interference of reflected acoustic waves.

To evaluate the performance of the applied denoising techniques, several spatial, edge-preserving, and transform-domain filters were applied to each dataset. The quality of the denoised images was quantitatively evaluated using widely adopted image quality metrics including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Signal-to-Noise

Ratio (SNR). These metrics provide complementary insights into denoising performance by measuring reconstruction error, signal fidelity, and perceptual structural similarity.

6.1 ANALYSIS OF SPATIAL DOMAIN FILTERS

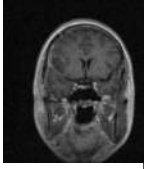
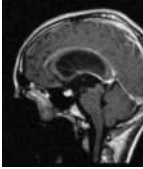
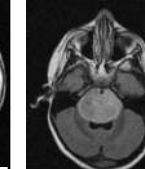
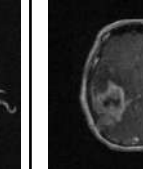
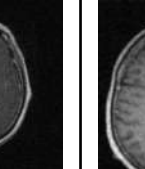
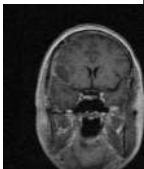
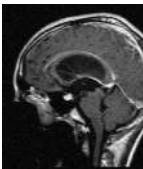
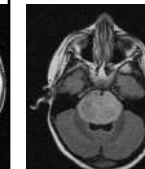
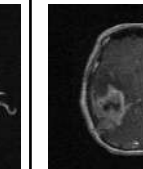
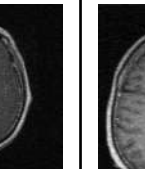
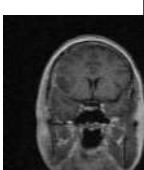
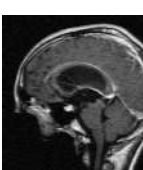
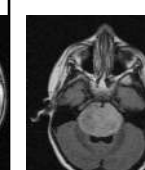
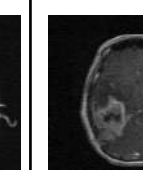
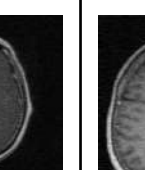
The Mean filter and Moving Average filter perform linear smoothing by averaging pixel values within a local neighborhood. While these filters effectively reduce random noise, they tend to blur edges and reduce fine structural details in medical images. This effect is particularly noticeable in CT and MRI images where preservation of anatomical boundaries is critical.

The Gaussian filter performs weighted smoothing using a Gaussian kernel, which allows stronger influence from nearby pixels while reducing the effect of distant ones. This approach generally produces smoother results than simple averaging filters and provides improved noise suppression while maintaining overall image structure.

The Median filter is a nonlinear filtering technique that replaces each pixel with the median value of its neighborhood. Unlike linear filters, the Median filter is highly effective in removing impulse noise while preserving edge information. This characteristic makes it particularly suitable for medical images where structural boundaries must be maintained.

The Hybrid filter combines the advantages of linear and nonlinear filtering approaches. By integrating mean-based smoothing with median-based edge preservation, the hybrid approach attempts to achieve a balance between noise suppression and structural preservation. In many cases, this filter improves structural similarity compared to simple smoothing filters.

Overall, spatial domain filters provide computationally efficient denoising and serve as useful baseline techniques . However, they may struggle to preserve subtle anatomical features when noise levels are high.

FILTER	Image1	Image2	Image3	Image4	Image5
Mean					
Moving Average					
Exponential Moving Average					

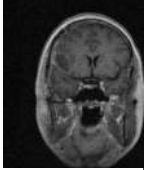
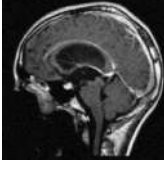
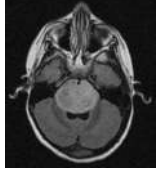
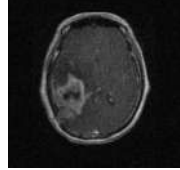
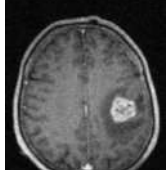
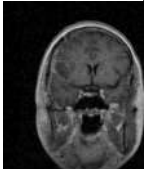
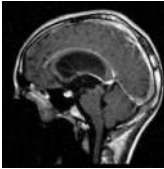
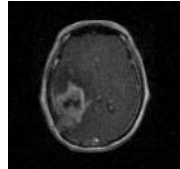
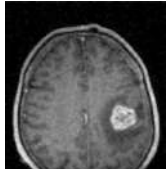
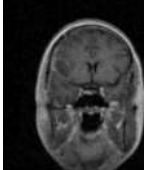
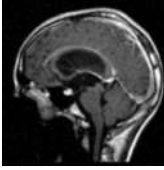
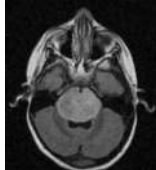
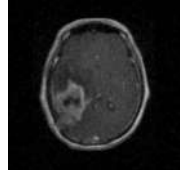
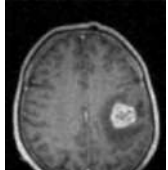
Gaussian					
Median					
Hybrid(Mean +Median)					

Fig 3: Visual comparison of MRI images after applying spatial domain filters

Table 2: Metrics for MRI images after applying spatial domain filters

FILTER	MSE	PSNR	SNR	SSIM
Mean	108.9268	28.2329	16.6734	0.5974
Moving Average	150.9043	26.4059	14.8464	0.4486
Exponential Moving Average	280.8526	23.9107	12.3511	0.4110
Gaussian	92.7836	28.7824	17.2228	0.6023
Median	69.8764	30.0223	18.4628	0.7110
Hybrid(Mean +Median)	96.7328	29.1379	17.5783	0.7601

FILTER	Image1	Image2	Image3	Image4	Image5
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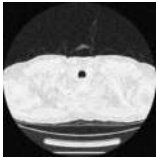
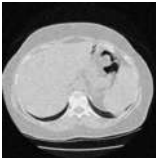
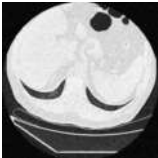
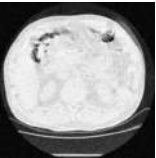
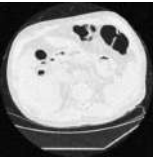
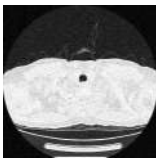
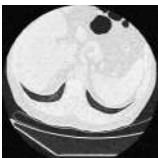
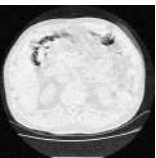
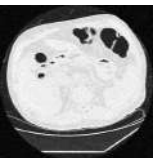
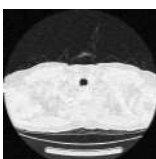
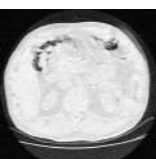
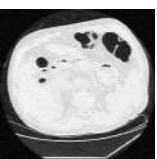
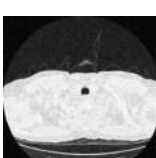
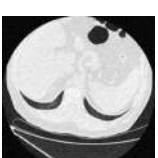
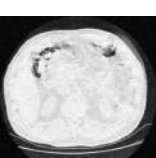
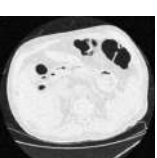
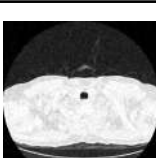
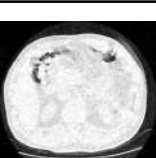
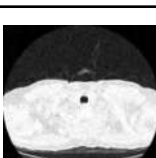
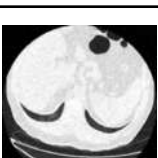
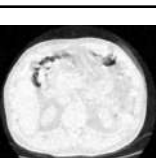
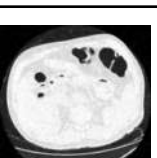
Mean					
Moving Average					
Exponential Moving Average					
Gaussian					
Median					
Hybrid(Mean+Median)					

Fig 4: Visual comparison of CT images after applying spatial domain filters

Table 3: Metrics for CT images after applying spatial domain filters

FILTER	MSE	PSNR	SNR	SSIM
Mean	487.6832	21.3456	17.6724	0.3016
Moving Average	322.6276	23.1911	19.5178	0.3988
Exponential Moving Average	423.5661	21.9744	18.3011	0.4354
Gaussian	394.8692	22.2866	18.6133	0.3573

Median	358.3133	22.6979	19.0246	0.4035
Hybrid(Mean+Median)	474.5965	21.4512	17.7780	0.4099

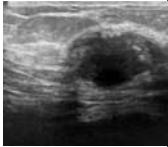
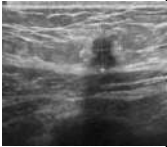
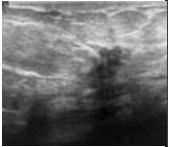
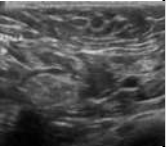
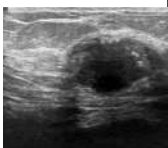
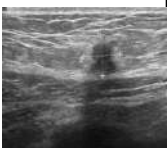
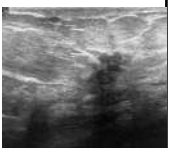
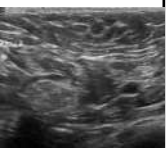
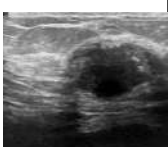
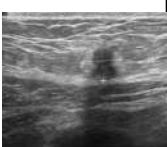
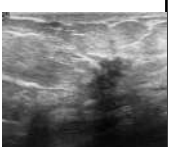
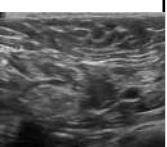
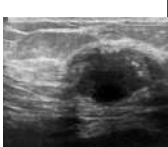
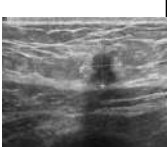
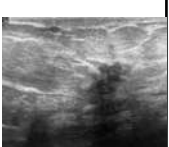
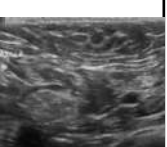
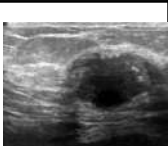
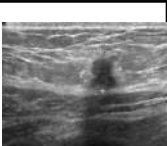
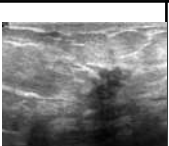
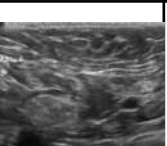
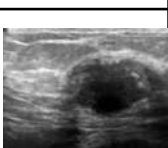
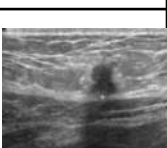
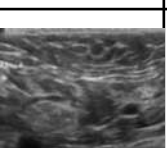
FILTER	Image1	Image2	Image3	Image4	Image5
Mean					
Moving Average					
Exponential Moving Average					
Gaussian					
Median					
Hybrid(Mean+Median)					

Fig 5: Visual comparison of ULTRASOUND images after applying spatial domain filters

Table 4: Metrics for ULTRASOUND images after applying spatial domain filters

FILTER	MSE	PSNR	SNR	SSIM
Mean	87.6084	28.7610	21.0393	0.8124

Moving Average	122.7506	27.3399	19.6181	0.7674
Exponential Moving Average	146.5179	26.5091	0.7308	18.7874
Gaussian	394.8692	22.2866	0.3573	18.6133
Median	358.3133	22.6979	0.4035	19.0246
Hybrid(Mean +Median)	474.5965	21.4512	0.4099	17.7780

For MRI images, the Median filter achieves the best quantitative performance with lowest MSE (69.8764) and highest PSNR (30.0223) and SNR (18.4628), while the Hybrid filter provides the best structural preservation with SSIM (0.7601). For CT images, the Moving Average filter performs optimally with MSE (322.6276), PSNR (23.1911), and SNR (19.5178), whereas the Hybrid filter slightly improves SSIM (0.4099). In Ultrasound images, the Mean filter outperforms others across all metrics, achieving MSE (87.6084), PSNR (28.7610), SNR (21.0393), and the highest SSIM (0.8124).

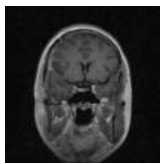
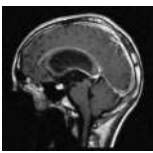
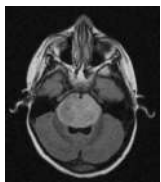
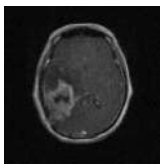
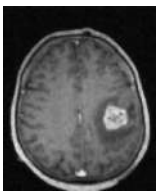
6.2 ANALYSIS OF EDGE PRESERVING FILTERS

The Bilateral filter performs edge-preserving smoothing by combining spatial proximity and intensity similarity. This dual weighting mechanism allows the filter to smooth homogeneous regions while preventing excessive smoothing across edges. As a result, the Bilateral filter often produces visually improved results compared to traditional spatial filters.

Anisotropic diffusion is an iterative filtering technique that selectively smooths image regions while preserving edges. The diffusion process is guided by gradient information so that smoothing occurs within homogeneous regions but is inhibited across strong intensity gradients. This method is particularly effective for preserving structural features in medical images.

The Non-Local Means (NLM) filter exploits self-similarity within the image by comparing pixel neighborhoods across a larger search region. Instead of relying solely on local information, the NLM filter averages pixels with similar patterns throughout the image. This approach can significantly improve denoising performance while preserving textures and structural details. However, the computational complexity of NLM is generally higher than traditional filtering methods.

Overall, edge-preserving filters demonstrate improved structural preservation compared to basic spatial filters, making them well-suited for medical image denoising tasks.

FILTER	Image1	Image2	Image3	Image4	Image5
Bilateral					

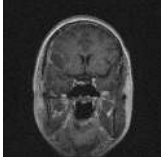
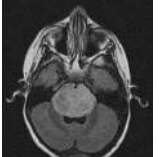
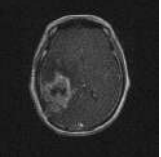
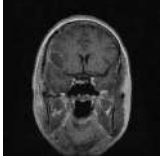
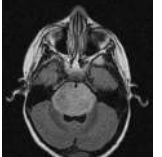
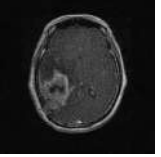
Anisotropic					
Non-Local Means					

Fig 6: Visual comparison of MRI images after applying edge preserving filters

Table 5: Metrics for MRI images after applying edge preserving filters

FILTER	MSE	PSNR	SNR	SSIM
Bilateral	101.8537	28.3259	16.7663	0.6007
Anisotropic	1999.5718	15.1295	3.5699	0.1191
Non-Local Means	374.3960	22.4392	10.8796	0.3196

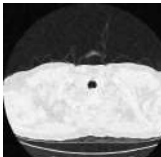
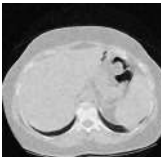
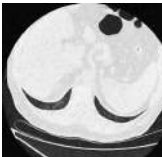
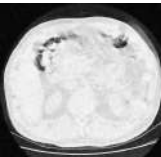
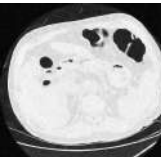
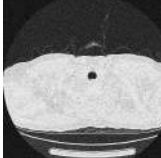
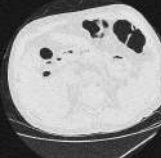
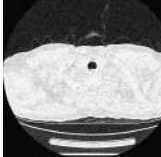
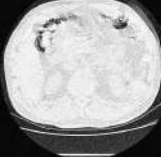
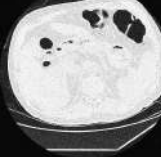
FILTER	Image1	Image2	Image3	Image4	Image5
Bilateral					
Anisotropic					
Non-Local Means					

Fig 7: Visual comparison of CT images after applying edge preserving filters

Table 6: Metrics for CT images after applying edge preserving filters

FILTER	MSE	PSNR	SNR	SSIM
Bilateral	234.1513	24.5306	20.8573	0.4204
Anisotropic	1681.2194	15.8916	12.2183	0.2901
Non-Local Means	407.9758	22.0292	18.3559	0.4616

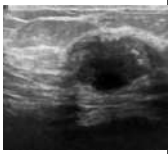
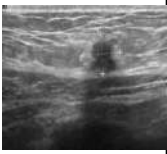
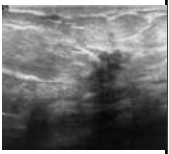
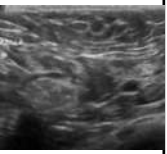
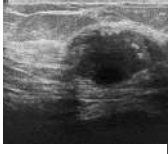
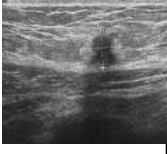
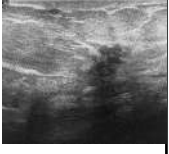
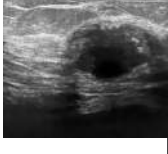
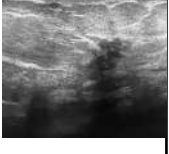
FILTER	Image1	Image2	Image3	Image4	Image5
Bilateral					
Anisotropic					
Non-Local Means					

Fig 8: Visual comparison of Ultrasound images after applying edge preserving filters

Table 7: Metrics for Ultrasound images after applying edge preserving filters

FILTER	MSE	PSNR	SNR	SSIM
Bilateral	94.1039	28.4429	20.7211	0.7730
Anisotropic	2229.9798	14.6592	6.9374	0.1379
Non-Local Means	496.9549	21.30168	13.5799	0.5643

The results indicate that among edge-preserving filters, the Bilateral filter consistently performs best across all modalities. For MRI, it achieves the lowest MSE (101.8537) and highest PSNR (28.3259), SNR (16.7663), and SSIM (0.6007). For CT, it again shows strong performance with MSE (234.1513), PSNR (24.5306), and SNR (20.8573), although Non-Local Means slightly improves SSIM (0.4616). In Ultrasound, the Bilateral filter outperforms others with MSE (94.1039), PSNR (28.4429), SNR (20.7211), and SSIM (0.7730).

In contrast, Anisotropic diffusion performs poorly across all datasets, with very high MSE (up to 2229.9798) and low PSNR, SNR, and SSIM, indicating loss of structural information.

6.3 ANALYSIS OF FREQUENCY AND TRANSFORM BASED METHODS

The Wiener filter is a statistical filtering technique that minimizes the mean squared error between the original and restored images. It adapts the filtering process based on local variance, allowing better noise suppression in regions with different noise characteristics.

Wavelet transform-based denoising operates by decomposing the image into multiple frequency subbands. Noise is typically concentrated in high-frequency components, which allows selective thresholding of wavelet coefficients to remove noise while preserving important structural information. Wavelet-based methods are widely used in medical image processing due to their ability to capture multi-resolution features.

Curvelet transform extends the capabilities of wavelet transforms by providing improved representation of edges and curved structures. Since many anatomical features in medical images exhibit curved boundaries, curvelet transforms can capture these structures more efficiently than traditional wavelet methods. This property often leads to improved structural preservation in denoised images.

Overall, transform-based techniques generally achieve better preservation of fine structural details compared to spatial filtering approaches, although they may require higher computational resources.

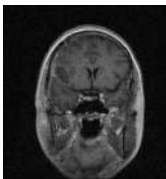
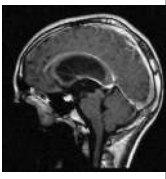
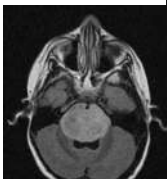
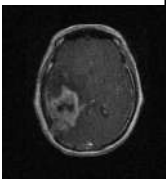
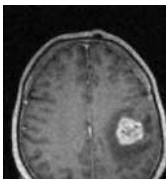
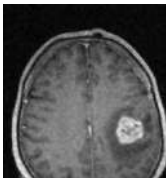
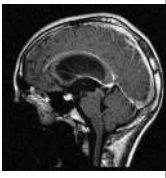
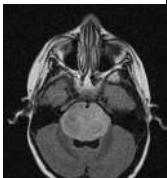
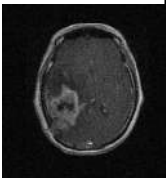
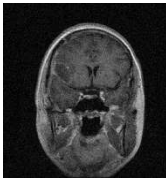
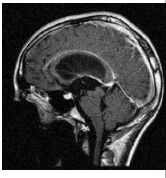
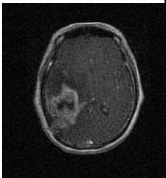
FILTER	Image1	Image2	Image3	Image4	Image5
Weiner					
Wavelet					
CurveLet					

Fig 9: Visual comparison of MRI images after applying frequency and transform based filters

Table 8: Metrics for MRI images after applying after applying frequency and transform based filters

FILTER	MSE	PSNR	SNR	SSIM
Weiner	108.5705	27.8645	16.3049	0.5665

Wavelet	304.0011	23.3131	11.7535	0.3181
CurveLet	475.3059	21.3701	9.8105	0.2551

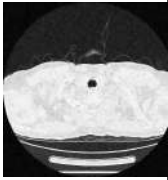
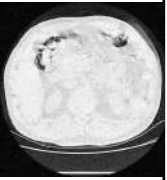
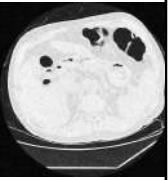
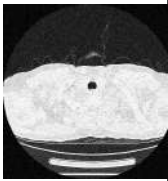
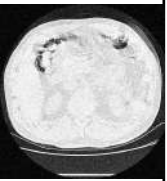
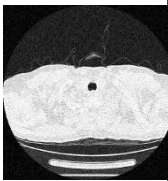
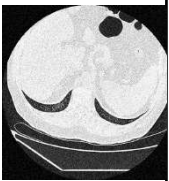
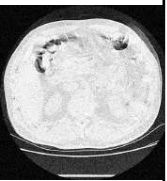
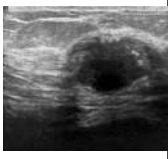
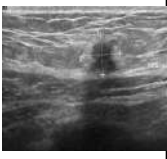
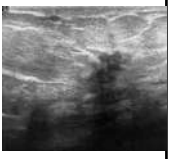
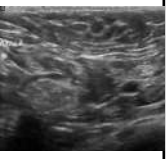
FILTER	Image1	Image2	Image3	Image4	Image5
Weiner					
Wavelet					
CurveLet					

Fig 10: Visual comparison of CT images after applying frequency and transform based filters

Table 9: Metrics for CT images after applying after applying frequency and transform based filters

FILTER	MSE	PSNR	SNR	SSIM
Weiner	253.4505	24.2345	20.5612	0.3670
Wavelet	327.8870	22.9753	19.3020	0.4797
CurveLet	463.7230	21.4737	17.8004	0.4435

FILTER	Image1	Image2	Image3	Image4	Image5
Weiner					

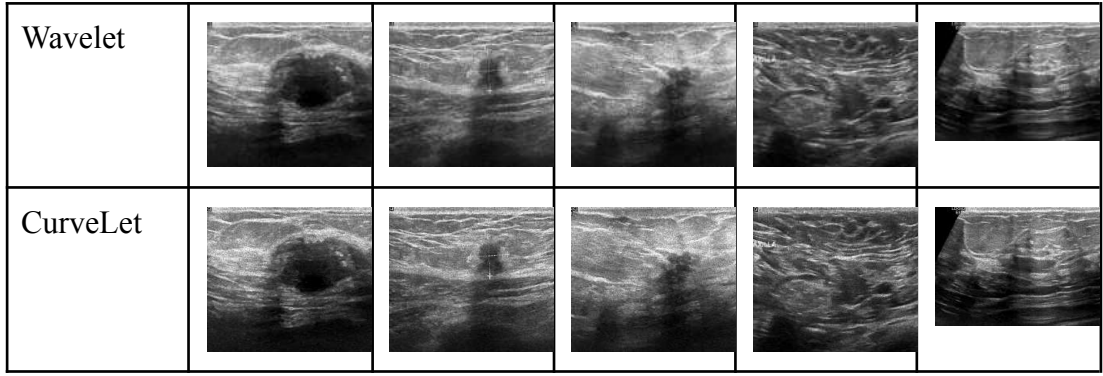


Fig 11: Visual comparison of Ultrasound images after applying frequency and transform based filters

Table 10: Metrics for Ultrasound images after applying after applying frequency and transform based filters

FILTER	MSE	PSNR	SNR	SSIM
Weiner	150.5482	26.3982	18.6765	0.7583
Wavelet	359.3280	22.7129	14.9912	0.5974
CurveLet	527.4858	21.0163	13.2945	0.4981

The results show that the Wiener filter consistently outperforms other transform-based methods across all modalities. For MRI, it achieves the best performance with lowest MSE (108.5705), highest PSNR (27.8645), SNR (16.3049), and SSIM (0.5665). For CT, Wiener again provides the lowest MSE (253.4505) and highest PSNR (24.2345) and SNR (20.5612), although Wavelet slightly improves SSIM (0.4797). In Ultrasound, Wiener remains superior with MSE (150.5482), PSNR (26.3982), SNR (18.6765), and SSIM (0.7583).

In contrast, Curvelet shows the weakest performance with the highest MSE (up to 527.4858) and lowest PSNR, while Wavelet provides moderate results with better structural similarity in some cases.

6.4 GRAPHICAL ANALYSIS OF EVALUATION METRICS ACROSS VARIOUS MODALITIES

The graphical analysis of MSE, PSNR, and SNR across CT, MRI, and Ultrasound images provides a clear comparative understanding of filter performance.. The MSE plots indicate that Anisotropic diffusion consistently produces the highest error values, confirming its poor denoising capability, while filters such as Bilateral, Median, and Wiener exhibit significantly lower MSE, indicating effective noise suppression.

The PSNR graphs demonstrate that Bilateral, Gaussian, Median, and Wiener filters achieve higher PSNR values, reflecting better reconstruction quality. In contrast, Curvelet and Anisotropic filters show lower PSNR, indicating degraded image quality.

Similarly, the SNR analysis highlights that Bilateral and Wiener filters maintain higher signal strength across all modalities, while Anisotropic diffusion shows the lowest SNR, indicating poor signal preservation. Overall, the graphical comparison reinforces that Bilateral and Wiener filters provide the most consistent and reliable performance across multiple metrics, while Anisotropic diffusion

underperforms, and Wavelet/Curvelet offer moderate results depending on the modality.

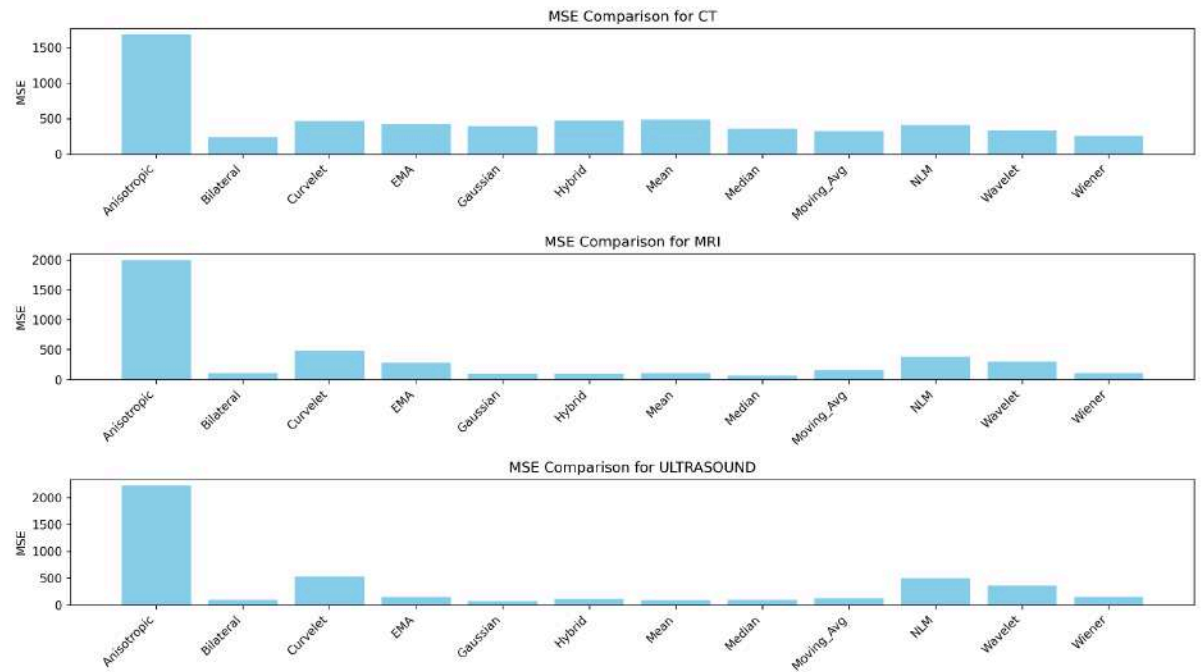


Fig 12: Graphical Analysis of MSE for Images of different modality after applying various denoising filters

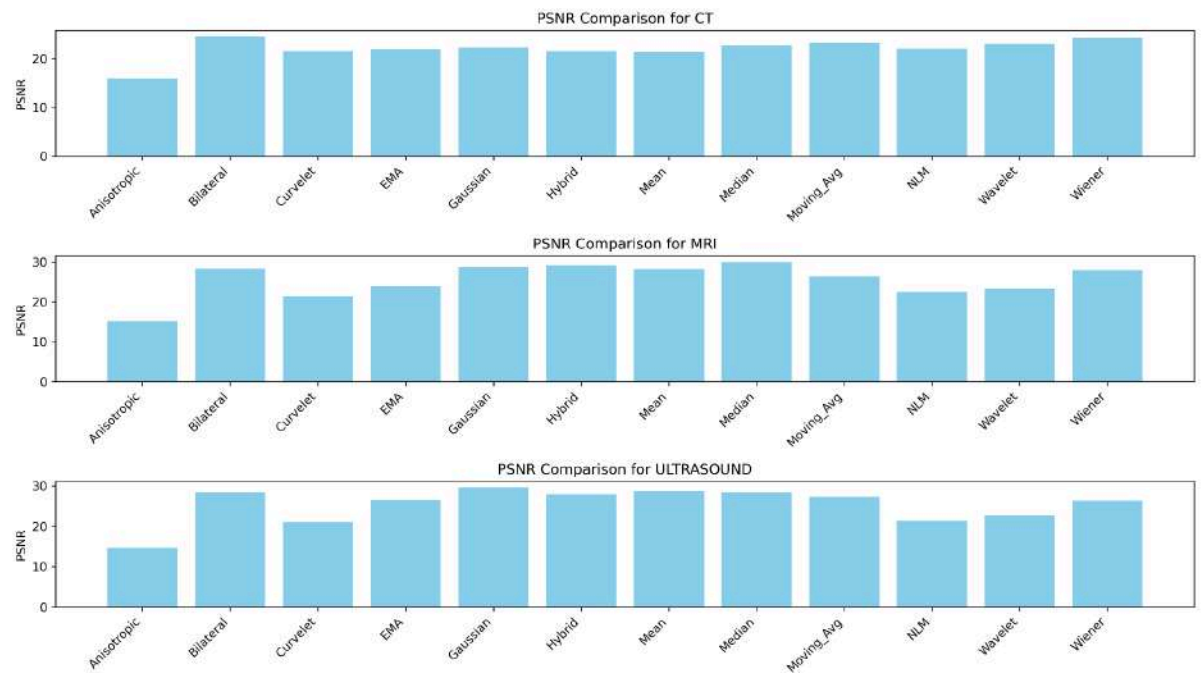


Fig 13: Graphical Analysis of PSNR for Images of different modality after applying various denoising filters

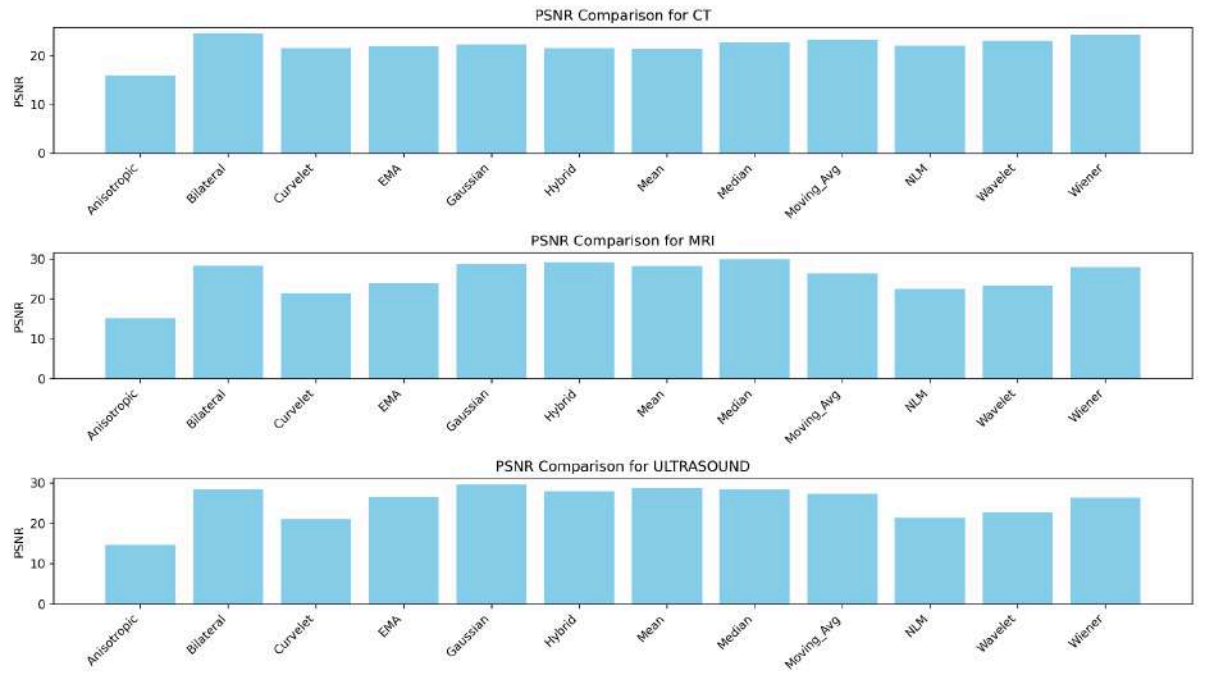


Fig 14: Graphical Analysis of SNR for Images of different modality after applying various denoising filters

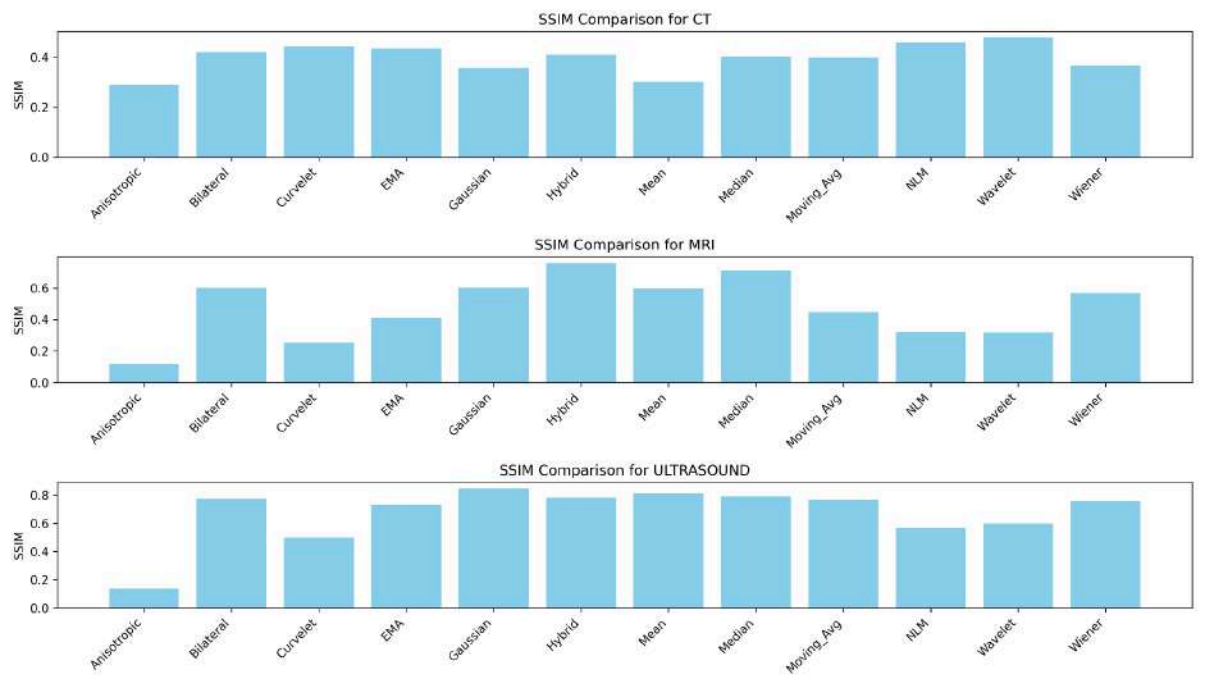


Fig 15: Graphical Analysis of SSIM for Images of different modality after applying various denoising filters

The numerical evaluation demonstrates that the Median filter achieves the best performance for MRI images, with $MSE = 69.8764$, $PSNR = 30.0223$, $SNR = 18.4628$, and $SSIM = 0.7601$, indicating superior noise reduction and structural preservation. For CT images, the Moving Average filter performs optimally, achieving $MSE = 322.6276$, $PSNR = 23.1911$, $SNR = 19.5178$, and $SSIM = 0.3988$, while for Ultrasound images, the Mean filter shows the best results with $MSE = 87.6084$, $PSNR = 28.7610$, $SNR = 21.0393$, and $SSIM = 0.8124$. Among advanced techniques, the Bilateral

filter provides strong edge-preserving performance, with values such as SSIM = 0.6007 (MRI) and 0.7730 (Ultrasound), whereas the Wiener filter emerges as the most effective transform-based method, achieving PSNR = 24.2345 (CT) and SSIM = 0.7583 (Ultrasound). In contrast, Anisotropic diffusion exhibits the poorest performance, with extremely high MSE values (up to 2229.9798) and low SSIM (as low as 0.1191). Overall, these results confirm that filter performance is highly dependent on the imaging modality, with different filters excelling across different datasets and evaluation metrics.

7. Applications in Medical Domain

Denoising significantly improves:

- Tumor segmentation in MRI
- Skull fracture detection in CT
- Fetal imaging clarity in ultrasound
- Retinal vessel enhancement
- 3D reconstruction and implant generation

Improved image quality enhances both manual diagnosis and automated machine learning pipelines.

8. Conclusion

This study presented a comprehensive comparative analysis of multiple denoising techniques applied to different medical imaging modalities, including Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and Ultrasound images. The evaluated techniques were categorized into three major groups: spatial domain filters, edge-preserving filters, and frequency/transform-based methods. The performance of each method was quantitatively assessed using widely accepted image quality metrics such as Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Signal-to-Noise Ratio (SNR).

The experimental results demonstrate that the performance of denoising algorithms strongly depends on the characteristics of the imaging modality and the type of noise present in the image. Spatial domain filters such as Mean, Gaussian, and Median filters provide computationally efficient noise reduction and are suitable for basic smoothing tasks. However, these methods may lead to edge blurring and loss of important structural information in medical images.

Edge-preserving filters such as Bilateral filtering, Anisotropic diffusion, and Non-Local Means improve the preservation of structural features and anatomical boundaries while reducing noise. These techniques are particularly useful for maintaining diagnostically important details in medical imaging.

Transform-domain approaches, including Wavelet and Curvelet transforms, demonstrated strong capabilities in capturing multi-scale image features and preserving structural details. These methods provide an effective balance between noise suppression and structural preservation, making them suitable for advanced medical image denoising tasks.

The quantitative evaluation demonstrates that denoising performance varies across modalities and techniques. For MRI images, the Median filter achieved the best results with MSE of 69.8764, PSNR of 30.0223 dB, and SNR of 18.4628, while the Hybrid filter provided the highest SSIM of 0.7601. In CT images, the Moving Average filter showed optimal performance with MSE of 322.6276, PSNR of 23.1911 dB, and SNR of 19.5178, whereas the Hybrid filter achieved the highest SSIM of 0.4099. For Ultrasound images, the Mean filter outperformed others with MSE of 87.6084, PSNR of 28.7610 dB, SNR of 21.0393, and SSIM of 0.8124. Among edge-preserving methods, the Bilateral filter consistently performed best, achieving, for example, MRI PSNR of 28.3259 dB and SSIM of 0.6007, and CT PSNR of 24.5306 dB. In transform-domain approaches, the Wiener filter yielded superior results across all modalities, with MRI PSNR of 27.8645 dB, CT PSNR of 24.2345 dB, and Ultrasound PSNR of 26.3982 dB, confirming its effectiveness in balancing noise suppression and structural preservation.

Overall, the comparative analysis highlights that no single denoising technique performs optimally for all imaging modalities. Instead, the selection of an appropriate denoising method should be guided by the noise characteristics and structural requirements of the specific medical imaging modality. The

findings of this study contribute to improved understanding of denoising techniques and support the development of more reliable medical image processing systems for diagnostic and clinical applications.

9. Future Work

Although the present study provides a comprehensive comparative analysis of several classical denoising techniques for CT, MRI, and Ultrasound images, there are several promising directions for further improvement and exploration.

One potential area for future research is adaptive parameter tuning. Many filtering techniques rely on manually selected parameters such as kernel size, diffusion coefficients, and threshold values. Developing adaptive parameter selection mechanisms that automatically adjust these parameters based on image characteristics and noise levels.

Future work may also involve comparative evaluation with deep learning-based denoising methods. Recent deep learning architectures such as DnCNN and U-Net have demonstrated strong performance in image restoration tasks. A detailed comparison between traditional filtering techniques and deep learning-based approaches would provide deeper insights into their relative advantages in medical imaging applications.

Additionally, extending the current work to 3D volumetric denoising represents an important research direction. Many medical imaging modalities such as CT and MRI generate volumetric datasets consisting of multiple slices. Applying denoising techniques directly on 3D volumes could help preserve spatial continuity between slices and improve the quality of reconstructed anatomical structures.

These future directions have the potential to further enhance the robustness, accuracy, and clinical applicability of medical image denoising techniques.

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